

# Benchmark RADAR: Living Search Engine for Retrieval and Discovery of AI Benchmark Research

Koutian Wu<sup>1,2 \*</sup> Junjie Zhou<sup>4</sup> Ergan Shang<sup>3</sup> Jiayu Wang<sup>5</sup> Pengqian Han<sup>6</sup>

<sup>1</sup> Earth and Space AI <sup>2</sup> Tacite AI <sup>3</sup> Carnegie Mellon University

<sup>4</sup> Hangzhou Dianzi University <sup>5</sup> Xi'an Jiaotong University

<sup>6</sup> The University of Auckland

## Abstract

Choosing an AI benchmark requires finding the test, its released artifacts, and evidence of how models perform on it. These records are scattered across papers, repositories, datasets, leaderboards, and model reports. Benchmark Radar brings them into a searchable catalog with source links, benchmark mentions, and evaluation settings attached to scores. At 2026-09-06, the catalog contained 1,283 benchmark records across four sources; 37 model reports supplied mentions of 110 benchmarks and 322 scores across 86 benchmark tracks. The reporting evidence reveals a concentrated set of widely used evaluations: GPQA Diamond appeared in 27 of the 37 reports. Eight benchmark tracks had a recorded score within five points of a 100-point ceiling, while six had later mentions at least 180 days after their latest archived score. These findings help readers identify evaluations with little remaining score headroom and gaps in score coverage. A worked prior-art search illustrates how a contributor used the catalog to assemble related work before designing a new evaluation. The system is available through a web dashboard, feeds, downloads, and local search clients.

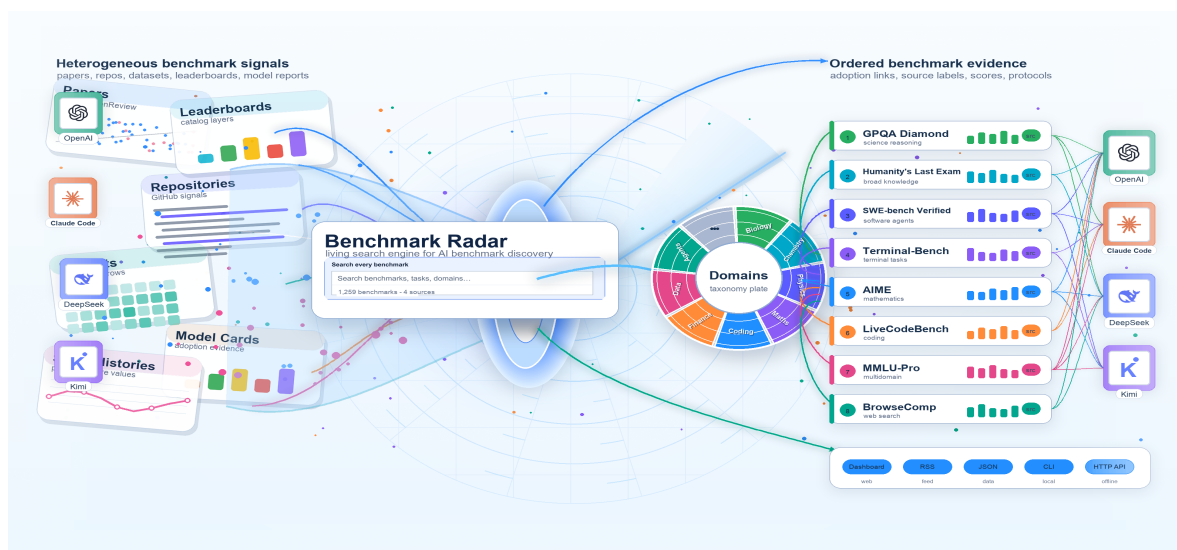


Figure 1: Benchmark Radar collects papers, repositories, datasets, and model reports, organizes them by topic, and makes them searchable alongside benchmark mentions and reported scores.

\*\*Corresponding author: [k@tacite.ai](mailto:k@tacite.ai)

# 1 Introduction

Before designing a new evaluation, a researcher needs to know which benchmarks already test the proposed capability, whether their data and code are available, and which models have been evaluated on them. Answering these questions requires following evidence across paper servers, code repositories, dataset hubs, leaderboards, and model reports. A benchmark name alone rarely supplies the test version, released artifacts, and evaluation settings needed to judge whether it fits the research question.

The range of evaluation tasks makes this search difficult. MMLU [11], GPQA [34], and Humanity’s Last Exam [30, 43] test knowledge and reasoning. SWE-bench tests issue resolution [16]; LiveCodeBench uses date-based code-generation tests [15]; and Terminal-Bench and Long-Horizon-Terminal-Bench evaluate command-line agents [21, 24]. Researchers must also find domain-specific tests, including FinanceBench [14], MATH [12], SciBench [36], LAB-Bench [20], and ChemBench [25]. Choosing among them requires inspecting tasks, scoring rules, and released artifacts.

LLM Stats, OpenCompass, and Artificial Analysis provide benchmark catalogs, evaluation platforms, and model comparisons [3, 22, 28]. Benchmark Radar builds on their records by adding daily discovery and model-report evidence to a shared search interface. A reader can retrieve a benchmark, follow its paper and artifact links, and inspect which reports mention it and under which settings they report scores. Each source record keeps its identity and citations so the reader can trace a result back to the evidence.

We present the collection and search methods behind this catalog and examine what the reporting evidence reveals. At the cutoff, GPQA Diamond appeared in reports from 11 organizations, eight benchmark tracks had recorded scores within five points of their ceiling, and six had a gap of at least 180 days between their latest score and a later mention. These observations connect benchmark selection to adoption, score headroom, and missing measurements. Appendix B follows a contributor who used Benchmark Radar to gather prior art for a proposed evaluation.<sup>1</sup>

## 2 Methodology

### 2.1 System Overview

Benchmark Radar adapts the daily collection approach of BuilderPulse, which answers a fixed question set from more than ten sources each morning [39]. It applies this approach to benchmark discovery, catalog search, and score histories. Readers can look for recent releases, search for a task or benchmark name, inspect mentions in model reports, and compare scores collected under matching evaluation settings. Table 1 lists these uses and the interfaces that support them.

Benchmark Radar converts source records into common fields for names, descriptions, identifiers, dates, and supporting documents. A source observation is a collected record; an artifact is a paper, repository, dataset, release, or page identified from those observations. The benchmark catalog retains one record per benchmark within each source, including records with missing measurements. Model reports describe model development and evaluation [26]. We measure adoption by documented benchmark mentions and record numeric scores separately.

### 2.2 Collection and Source Health

Connectors are programs that retrieve records from public sources. Each collection run searches a 48-hour window, records counts and errors by source, removes future-dated rows, and requires healthy

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<sup>1</sup>Source code: <https://github.com/ktwu01/benchmark-radar>.

| Interface      | Reader question                                 | Role in Benchmark Radar                                                                   |
|----------------|-------------------------------------------------|-------------------------------------------------------------------------------------------|
| Today          | What appeared or changed?                       | Ranks new records and links the source evidence.                                          |
| Search box     | Which records match this name, task, or domain? | Matches names, phrases, and query words in the benchmark index.                           |
| Leaderboard    | Which benchmarks do labs report?                | Counts benchmark mentions across curated model-card documents.                            |
| Scores         | Which reported scores can be compared?          | Groups scores by benchmark version, metric, and evaluation settings.                      |
| Trends and map | Which topics and sources recur?                 | Records source coverage so changes in collection can be considered when comparing topics. |
| CLI and HTTP   | Can an analyst reproduce search locally?        | Use one query service and a shared JSON response format.                                  |

Table 1: Reader questions and the interfaces used to answer them.

responses from a configured set of core sources before publication. The core sources at the cutoff were arXiv [4], Hugging Face Hub [13], and GitHub Search [9]. Optional routes include GitHub organizations, Hugging Face Papers, Kaggle datasets [17], Zenodo [42], OpenReview [29], Semantic Scholar [19], GitHub Releases [8], OpenAlex [31], Brave Search [5], 24 institution-operated research or engineering feeds, and a separate collector of public attention through Hacker News search on Algolia [1]. All three core sources were healthy at the cutoff. Semantic Scholar was rate-limited, Brave Search lacked an API key, and OpenReview returned no eligible rows. Appendix A lists the healthy connectors and monitored feeds.

The collection run at the cutoff fetched 867 rows, reduced them to 837 candidate records after removing duplicates, classified 322 as eligible, and recommended 110 for readers to review. Two collections ran within the cutoff window; after merging, the daily page contained 322 records, including 110 recommendations. Published records retain the source URL, retrieval time, parser version, source identifier, timestamps, authors or organizations when supplied, and a SHA-256 checksum for verifying the stored record. Raw responses and credentials are excluded.



*Dated source records and fixed collection settings make the results reproducible.*

Figure 2: Collection and publication stages. Connectors retrieve public records and check source health and record formats. Matching DOI, arXiv, OpenReview, GitHub, or Hugging Face identifiers link observations to artifacts. Classification and priority scoring organize the records before publication. Web pages, feeds, and local clients expose the collected evidence.

Dated snapshots, record formats, parser versions, and checksums allow the published data to be rebuilt and checked. Publication requires healthy core sources; failures in optional sources remain visible. Installed clients validate an update before activating it. Identity resolution relies on exact identifiers and reviewed links between source records. The evaluated release passed 1,236 tests after a clean rebuild. Section 4 describes the remaining limitations.

## 2.3 Identity, Ranking, and Search

The resolver groups observations when they share a DOI, arXiv identifier, OpenReview identifier, GitHub repository, or Hugging Face artifact. Similar titles alone do not force a merge; they remain separate until review confirms that they describe the same artifact. This policy reduces false joins but leaves possible duplicates visible when papers, repositories, and catalogs lack a common identifier.

Search covers four catalog sources, as summarized in Table 2: LLM Stats, OpenCompass Hub, Artificial Analysis, and Model reports. The index contains 1,283 source records, including 110 model-report benchmarks. Of these model-report benchmarks, 86 have reported scores and 24 have no scores; both groups remain searchable. Search matches query words and phrases against record fields, including benchmark names. It searches all four sources together and preserves each result’s source label. The same query against the same index produces the same ranking.

A separate priority score for daily recommendations combines relevance (35%), evidence (20%), recency (20%), and adoption (25%) on a 0–100 scale. These weights describe daily reading priority; search relevance depends on the query. The scoring configuration is included with the published data.

| Catalog source      | Rows  | Primary use                                                                         |
|---------------------|-------|-------------------------------------------------------------------------------------|
| LLM Stats           | 687   | Benchmark names and descriptions with source provenance.                            |
| OpenCompass Hub     | 461   | Paper, repository, release, and dataset links with source-specific identity fields. |
| Artificial Analysis | 25    | Current commercial evaluation catalog entries.                                      |
| Model reports       | 110   | Benchmarks mentioned in model reports, including records without scores.            |
| Total               | 1,283 | All four sources, retaining each record’s source label.                             |

Table 2: Catalog sources in the Benchmark Radar query index.

## 2.4 Adoption and Score Curation

Benchmark Radar records benchmark mentions and numeric scores from model-card and system-description documents [26]. This collection includes 37 model documents from 12 organizations. They mention 110 benchmarks whose names have been reconciled within the model-report collection. When a document contains an interpretable numeric value and sufficient context, the score archive records the benchmark version and metric (the instrument), along with evaluation settings (the protocol). These settings include examples supplied in the prompt (shots), tool access, attempts, and the evaluator when available. The archive contains 322 scores across 86 benchmark tracks. A score track groups results for a named benchmark; comparisons within it still require matching instruments and protocols. Scores from benchmark registries remain outside these model-report score histories unless they can be connected to a comparable protocol [7, 26].

# 3 Results

## 3.1 Coverage at the Cutoff

At the 2026-09-06 cutoff, the cumulative discovery collection held 10,702 source observations and 6,304 artifacts grouped by exact identifiers. Primary or structured sources supplied 10,473 observations, or 97.86% of the collection. This category covers scholarly records, institution-operated feeds, code repositories, and dataset or artifact hosts. Discovery observations, benchmark records, model-report

mentions, and scores count different objects and must not be added together. Table 3 defines each unit.

|                                                             |                                                              |                                                                     |                                                            |
|-------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------|
| <b>10,702</b><br>source observations<br>across 45 snapshots | <b>6,304</b><br>artifacts grouped by<br>matching identifiers | <b>1,283</b><br>searchable benchmark<br>records, including unscored | <b>4</b><br>catalog sources<br>with source labels retained |
|-------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------|

**An observation is a collected source record. Matching identifiers link observations to artifacts. The catalog counts benchmark records separately for each source.**

*Data cutoff: 2026-09-06. Catalog records and discovery observations are separate units.*

Figure 3: Source observations, resolved artifacts, searchable benchmark records, and catalog sources as of 2026-09-06.

| Unit                | Count  | Interpretation                                                                       |
|---------------------|--------|--------------------------------------------------------------------------------------|
| Snapshot            | 45     | Dated collection records, including 41 observed and 4 simulated snapshots.           |
| Source observation  | 10,702 | Collected records with their source and retrieval information.                       |
| Unique artifact     | 6,304  | Papers, repositories, datasets, releases, or pages grouped by matching identifiers.  |
| Searchable entry    | 1,283  | One benchmark record per source, including records without scores.                   |
| Adoption benchmark  | 110    | Benchmarks mentioned in model reports, with names reconciled within that collection. |
| Model-card document | 37     | Curated reports from 12 organizations.                                               |
| Curated score       | 322    | Numeric results across 86 benchmark tracks.                                          |

Table 3: What each count measures at 2026-09-06.

### 3.2 Source Composition and Reliability

Five source labels account for 9,532 of 10,702 observations, or 89.1% of the cumulative collection. Because these sources supply most observations, their retrieval caps, rate limits, and availability can change the apparent topic mix. At the cutoff, the required sources were healthy, while optional routes varied in yield because of rate limits, missing credentials, or empty result windows.

Fifty-nine of 6,304 artifacts were linked across more than one source, or 0.94% of the artifact set. Matching exact identifiers reduces false joins; links between papers, code, and datasets still require review when identifiers are missing or inconsistent.

### 3.3 Relationship to Existing Catalogs

Benchmark Radar imports records from three benchmark registries: LLM Stats, OpenCompass Hub, and Artificial Analysis. These projects have different goals. LLM Stats publishes live benchmark rankings and benchmark-family pages [22]; OpenCompass is primarily an evaluation platform with a searchable dataset-statistics page [28]; and Artificial Analysis publishes model evaluations and a documented intelligence-benchmarking methodology [3].

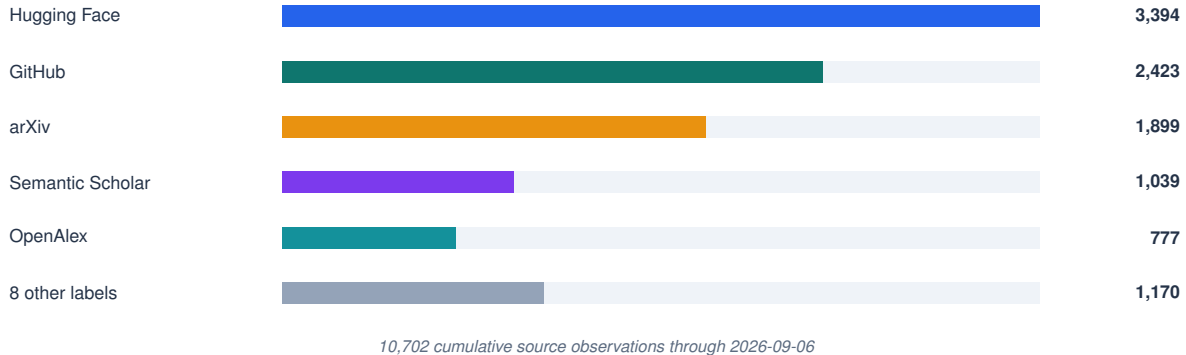


Figure 4: Discovery observations by source through 2026-09-06. The bars count collected observations, including repeat sightings, rather than distinct benchmarks.

The index contains 687 records from LLM Stats, 461 from OpenCompass Hub, 25 from Artificial Analysis, and 110 from Model reports, yielding 1,283 searchable source records. These counts describe each source’s contribution to the catalog. Benchmark Radar adds daily discovery, links between artifacts with matching identifiers, and model-report mentions and scores. Readers can search this combined evidence while inspecting the source and definition of each record.

### 3.4 Benchmark Adoption in Model Reports

In the model-report collection, eight benchmarks appeared in documents from at least six organizations: GPQA Diamond [34], Humanity’s Last Exam [30], SWE-bench Verified [16], Terminal-Bench [24], AIME [23], LiveCodeBench [15], MMLU-Pro [37], and BrowseComp [38]. GPQA Diamond appeared in 27 of 37 documents from 11 organizations, making it the most commonly mentioned benchmark in these reports. This concentration suggests that frontier-model reporting has partly converged on a common set of evaluations, even as researchers introduce benchmarks in other areas.

Score coverage lagged behind mentions for six benchmarks: GSM8K, HumanEval, MBPP, MMLU, DROP, and MATH-500. Each appeared in a model report at least 180 days after its latest archived numeric score; the observed gaps ranged from 343 to 775 days. Table 4 lists the dates. A reader investigating these benchmarks has newer reports to consult for score extraction.

| Benchmark | Latest score date | Latest mention date | Gap (days) |
|-----------|-------------------|---------------------|------------|
| GSM8K     | 2024-03-08        | 2026-04-22          | 775        |
| HumanEval | 2024-09-18        | 2026-04-22          | 581        |
| MMLU      | 2025-01-22        | 2026-04-22          | 455        |
| DROP      | 2025-01-22        | 2026-04-22          | 455        |
| MBPP      | 2024-09-18        | 2025-11-20          | 428        |
| MATH-500  | 2025-05-14        | 2026-04-22          | 343        |

Table 4: Benchmarks whose latest model-report mention follows their latest archived score by at least 180 days. Dates come from the score and model-report archives at the manuscript cutoff. The gap measures archive coverage.

### 3.5 Near-Ceiling Reported Scores

Eight of the 86 model-report benchmark tracks had a highest archived score within five points of a 100-point ceiling.<sup>2</sup> Table 5 retains the model and evaluation settings behind each value. Headroom is the metric’s upper bound minus the reported score. For example, GLM-5.2 scored 99.2 on AIME 2026, leaving 0.8 points, while Gemini 3.1 Pro scored 99.3 on the Telecom domain of tau2-bench, leaving 0.7 points. These results identify test settings where further gains have little room on the reported scale.

| Benchmark           | Model and source     | Score | Gap | Recorded settings                                                            |
|---------------------|----------------------|-------|-----|------------------------------------------------------------------------------|
| AIME 2026           | GLM-5.2 [40]         | 99.2  | 0.8 | Temperature 1.0; top- $p$ 0.95; 163,840-token limit; GPT-5.5 (medium) judge. |
| Arena-Hard          | Qwen3-235B-A22B [32] | 95.6  | 4.4 | Thinking mode; win rate.                                                     |
| DeepSearchQA        | Kimi K3 [27]         | 95.0  | 5.0 | F1; maximum reasoning effort.                                                |
| HMMT Nov. 2025      | GLM-5 [41]           | 96.9  | 3.1 | Temperature 1.0; top- $p$ 0.95; GPT-5.2 (medium) judge.                      |
| MATH-500            | Qwen3-235B-A22B [32] | 98.0  | 2.0 | Thinking mode.                                                               |
| MathVision          | Kimi K3 [27]         | 97.8  | 2.2 | Maximum reasoning effort; Python available.                                  |
| SWE-bench Verified  | Claude Opus 5 [2]    | 96.0  | 4.0 | Adaptive thinking; maximum effort; average of five trials.                   |
| tau2-bench, Telecom | Gemini 3.1 Pro [10]  | 99.3  | 0.7 | High thinking level; Telecom domain.                                         |

Table 5: Highest archived scores for the eight near-ceiling model-report tracks. Scores use 0–100 scales; gaps are percentage points below 100. The metrics include accuracy, pass@1, F1, resolved-task rate, and win rate. Each row links to its reporting source.

The test version and settings determine what each small gap means. AIME 2026 and AIME 2024 use different problem sets; the 99.2 result above and the archive’s older AIME 2024 result of 79.8 describe different tests. Likewise, the tau2-bench value above concerns Telecom, and the MathVision value includes Python access. Benchmark Radar preserves these details so readers can select the relevant test and inspect the score in context. Comparisons require checking the source documents for matching versions, reasoning budgets, tools, attempts, and evaluators [7, 26].

## 4 Limitations and Future Work

Lexical search can miss paraphrased benchmark names, renamed tasks, or descriptions of related ideas. Semantic retrieval could broaden the candidate set while preserving source links and retrieval explanations [33]. The worked example combines catalog queries with web search; evaluating retrieval quality requires relevance judgments over a broader set of queries and candidate records.

The collected evidence also has gaps. Public endpoints, credentials, and rate limits affect source coverage. Missing identifiers leave links between papers, code, and datasets unresolved. In the snapshot, 12,594 scores from benchmark registries remain outside the model-report score histories because they lack protocol fields such as prompt examples, tool access, attempts, evaluator, date-based test splits, or evaluation software. Matching archive labels alone cannot verify identical evaluation conditions or independent repeated runs. Further curation should capture these settings from primary reports and distinguish score-publication dates from evaluation dates.

<sup>2</sup>Junjie Zhou prepared the score-archive audit and report revisions underlying this analysis, documented in [issue #457](#).



Task-level coverage is incomplete. The capability-classification component, KW-Bench, has 5,661 tracks without complete task-capability labels at the cutoff. A classification track is an artifact or a declared task within a benchmark suite, drawn from discovery observations. Completing these labels would help readers search by the ability a task tests.

## 5 Conclusion

Benchmark Radar makes 1,283 benchmark records searchable alongside their sources, model-report mentions, and score histories. The reporting evidence identifies widely used tests, near-ceiling results, and benchmarks whose score coverage has fallen behind later mentions. Researchers can use these records to assemble prior art, inspect evaluation settings, and target missing evidence for review. The next priorities are broader retrieval evaluation, task-capability labels, and more complete protocol capture from primary reports.

*Author note.* Koutian Wu led the work and manuscript preparation, built the initial collection pipeline, and implemented data aggregation across sources. Ergan Shang prepared figures and contributed copyediting. Pengqian Han contributed copyediting.

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# Appendix

## A Source Inventory and Health

Table 6 lists source connectors that completed successfully at the cutoff. Row counts are measured before duplicate removal and ranking. A connector can be healthy with zero eligible rows if its request succeeds and the response can be parsed. Core sources cover papers, shared model or dataset artifacts, and code; all must be healthy before publication. Failures in optional sources are reported but do not block publication.

| Connector            | Role                                                                      | Cutoff status             | Core |
|----------------------|---------------------------------------------------------------------------|---------------------------|------|
| arXiv                | Primary papers via selected AI, language, vision, and software categories | Healthy; 0 rows           | Yes  |
| Hugging Face Hub     | Datasets and Spaces connected to benchmark artifacts                      | Healthy; 92 rows          | Yes  |
| GitHub Search        | Code repositories and benchmark artifacts                                 | Healthy; 300 rows; at cap | Yes  |
| GitHub Organizations | Reviewed organization repositories                                        | Healthy; 4 rows           | No   |
| Hugging Face Papers  | Community-surfaced papers                                                 | Healthy; 0 rows           | No   |
| Kaggle Datasets      | Public benchmark datasets                                                 | Healthy; 32 rows          | No   |
| Zenodo               | DOI-bearing research artifacts                                            | Healthy; 57 rows          | No   |
| OpenReview           | Conference submissions                                                    | Healthy; no eligible rows | No   |
| GitHub Releases      | Curated first-party releases                                              | Healthy; 6 rows           | No   |
| OpenAlex             | Scholarly discovery metadata                                              | Healthy; 200 rows         | No   |

Table 6: Healthy direct discovery connectors at the current cutoff.

Table 7 lists research and engineering feeds operated by the institutions being monitored. They can surface benchmark papers, datasets, model announcements, and engineering posts. The feed collector yielded one relevant record at the cutoff. Some healthy feeds produced zero matches after relevance filtering. Searches restricted to an organization’s website cover institutions that lack a verified feed.

| Feeds 1-12                      | Feeds 13-24      |
|---------------------------------|------------------|
| Meituan Engineering             | Ai2              |
| OpenAI News                     | Together AI      |
| Google AI                       | Sakana AI        |
| Google DeepMind                 | Qwen             |
| Google Research                 | Ollama           |
| Apple Machine Learning Research | Stability AI     |
| AWS Machine Learning            | Nomic AI         |
| Hugging Face Blog               | Replicate        |
| Microsoft Research              | NVIDIA Developer |
| NVIDIA AI Blog                  | IBM Research     |
| Mistral AI                      | Databricks       |
| Meta Research                   | LangChain        |

Table 7: First-party research and engineering feeds monitored by Benchmark Radar.

## B Worked Use Case: Prior-Art Check for a New Evaluation

The contributor wanted to determine whether a proposed evaluation duplicated existing work. During August 2026, they surveyed recent work on credit assignment in agentic training, using small Qwen-series models as a requirement for reproducible baseline experiments. The search produced a focused comparison table that the contributor used to assess the proposed design.

The contributor gave the task to a coding agent together with the public setup instructions for Benchmark Radar. The agent installed the command-line client and the Benchmark Radar Skill, a set of instructions for using the client. It then downloaded the local corpus and queried candidate records offline.

```

what can i do with benchmark-radar cli?

+ Thought: 328ms

$ benchmark-radar --help

usage: benchmark-radar [-h] [--author-output AUTHOR_OUTPUT]
                        [--author-contacts AUTHOR_CONTACTS]
                        [--author-repo-limit AUTHOR_REPO_LIMIT]
                        [--author-contributors AUTHOR_CONTRIBUTORS]
                        [--author-emails] [--config CONFIG] [--output OUTPUT]
                        [--json-output JSON_OUTPUT]
                        [--social-output SOCIAL_OUTPUT] [--items ITEMS]
                        [--repo REPO] [--channels CHANNELS]
                        [--existing-body EXISTING_BODY] [--since SINCE]
                        [--until UNTIL] [--snapshot-dir SNAPSHOT_DIR]

Click to expand

+ Thought: 531ms

The CLI has two groups:

Consumer (query) commands - for searching and inspecting local benchmark data:

- search "query" --scope catalog[radar|all] --json - discover benchmarks (catalog = verified records, radar = recent unverified evidence like papers/repos/datasets)
- show "key-or-slug" --json - inspect one record in detail
- recent --json - newest Radar evidence
- status --json - local data health/provenance
- init / sync - set up or refresh the local data
- serve --host 127.0.0.1 --port 8765 - local HTTP interface for the same queries

Maintainer commands - for operating the radar pipeline (not for ordinary use): run, rebuild, backfill, migrate, rescore, simulate-history, export, classify, authors, social, normalize-external, build-data-release.

For your local benchmark search use case, you'll mostly need search, show, recent, status, and serve.

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..... esc Interrupt 150.1K (15%) · $0.03 ctrl+p commands

```

Figure 5: A coding agent installs the Benchmark Radar client and its usage instructions, then runs local queries.

The agent inspected paper, repository, and dataset links to decide which sources to open next. Figures 6 and 7 show two retrieved records: one has a repository link, and the other has both paper and repository links. Neither records a dataset link, leaving dataset availability for a follow-up check.

```

    "rank": 18,
    "recommended": false,
    "score": 18.75,
    "slug": null,
    "snapshot_date": "2026-09-01",
    "source": "Hugging Face Papers",
    "source_id": "2608.28476",
    "updated_at": "2026-08-28T00:00:00+00:00",
    "url": "https://huggingface.co/papers/2608.28476"
  },
  {
    "categories": [
      "benchmark"
    ],
    "description": "LTABC-KM is a long-tail-aware adaptive artificial bee colony-K-means for unsupervised image clustering. It leverages density-aware coreset and graph-propagation assignment to recover minority tail modes without labels, achieving significant Macro-F1, NMI and ARI improvements on imbalanced benchmarks with favorable quality-computation trade-offs.",
    "has_dataset": false,
    "has_paper": false,
    "has_repo": true,
    "has_size": false,
    "key": "radar:github:Lutra11/LTABC-KM",
    "kind": "radar",
    "languages": [],
    "match": {
      "idf_coverage": 0.4391,
      "matched_fields": [
        "description"
      ],
      "matched_tokens": [
        "assignment"
      ],
      "missing_tokens": [
        "credit",
        "training"
      ],
      "phrase_fields": [],
      "query_coverage": 0.3333,
      "ranking_algorithm": "bm25f_v3",
      "reason": "1 of 3 unique query tokens matched across fields; IDF coverage 0.44",

```

"has\_dataset": false,  
 "has\_paper": false,  
 "has\_repo": true,  
 "has\_size": false,

```

    "key": "radar:github:Lutra11/LTABC-KM",
    "kind": "radar",
    "languages": [],
    "match": {
      "idf_coverage": 0.4391,
      "matched_fields": [
        "description"
      ],
      "matched_tokens": [
        "assignment"
      ],
      "missing_tokens": [
        "credit",
        "training"
      ],
      "phrase_fields": [],
      "query_coverage": 0.3333,
      "ranking_algorithm": "bm25f_v3",
      "reason": "1 of 3 unique query tokens matched across fields; IDF coverage 0.44",

```

□
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D:\projects\radar-test
187.3K (19%) · \$0.04 **ctrl+p** commands

Figure 6: A candidate record with a repository link for inspecting the implementation. Paper and dataset links are missing from the record.

```
"source": "GitHub",
"source_id": "mdkamranalam/credixa",
"updated_at": null,
"url": "https://github.com/mdkamranalam/credixa"
},
{
  "categories": [
    "benchmark"
  ],
  "description": "Long-horizon agentic tasks require large language models (LLMs) to iteratively retrieve, integrate, and maintain dispersed information across multi-turn interactions, but preserving all interaction histories leads to a continuously growing working context. Recent proactive context management methods allow models to edit their own working context with specialized tools, yet they still face three key limitations: (1) a limited toolset restricted to search, deletion, and summarization, with no support for global planning, long-term memory, and adaptive compression; (2) inefficient exploration that treats context management actions uniformly despite their heterogeneous impacts on final outcomes; and (3) coarse-grained credit assignment that assigns the final trajectory-level reward to all intermediate context editing actions during RL. To bridge these gaps, we introduce ContextPilot, a proactive context management framework for long-horizon agentic reasoning. Our approach systematically augments the toolset with planning, long-term memory, and soft context offloading tools. We further propose an RL method tailored for context management, which uses context and entropy variation to identify critical editing decisions for branch sampling and estimates action-level advantages from all branched trajectories that pass through the corresponding context editing action. Experiments on long-context QA and deep search tasks show that ContextPilot achieves stronger performance with a more compact working context, consistently outperforming existing baselines across various base models and benchmarks. Code is available at https://github.com/Tencent/ContextPilot.",
  "has_dataset": false,
  "has_paper": true,
  "has_repo": true,
  "has_size": false,
  "key": "radar:hugging face papers:2608.28476",
  "kind": "radar",
  "languages": [],
  "match": {
    "idf_coverage": 0.8392,
    "matched_fields": [
      "description"
    ],
    "matched_tokens": [
      "assignment",
      "credit"
    ]
  },
}
```

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D:\projects\radar-test187.3K (19%) · \$0.04ctrl+p commands

Figure 7: A second candidate provides both paper and repository links. Its dataset link needs a follow-up check.

The agent supplemented catalog queries with web search to find related work described in different terms. It cross-checked the candidate sets and inspected the source evidence before selecting records for the comparison. Figure 8 shows the word matches and score components available for inspecting one retrieved candidate.

```

"kind": "radar",
"languages": [],
...
Click to expand

$ benchmark-radar search "credit assignment training" --scope radar --json

...
"query_coverage": 0.3333,
"ranking_algorithm": "bm25f_v3",
"reason": "1 of 3 unique query tokens matched across fields; IDF coverage 0.44",
"retrieval_score": 10.824384,
"score_components": {
  "bm25f": 10.824384,
  "name_bonus": 0.0,
  "phrase_bonus": 0.0
},
"modality": null,
"name": "deepvisionaiofficial/Automation_AI-Assignment-Checker",
"openness": null,
"published_at": "2026-08-10T05:56:17+00:00",
"publisher": "",
"rank": 6,
"recommended": false,
"score": 36.32,
"slug": null,
"snapshot_date": "2026-08-10",
"source": "GitHub",
"source_id": "deepvisionaiofficial/Automation_AI-Assignment-Checker",
"updated_at": null,
"url": "https://github.com/deepvisionaiofficial/Automation_AI-Assignment-Checker"
},
{
  "categories": [
    "benchmark",
    "agentic"
  ]
}

```

use internal rubrics to compute retrieval score

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D:\projects\vradar-test 187.3K (10%) · \$0.04 ctrl+ip commands

Figure 8: A candidate record with query-word matches and retrieval-score components for the agent to inspect when judging relevance.

Figure 9 shows the resulting related-work table, which the contributor judged complete enough to act on.

| # | Work                                                                               | Published  | Qwen base model(s)                                                         | Credit-assignment focus                                                                                              | Agentic benchmarks used                                                | Link                                                                                                                                                                            |
|---|------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | SRPO: Self-Reflective Policy Optimization for Long-Horizon Reasoning               | 2026-08-25 | Qwen3-8B (also Qwen3-1.7B / 32B scaling)                                   | Reflection-conditioned dense token-level signals convert sparse terminal reward into training signal                 | AIIME24, WebShop, ALFWorld, SWE-Bench-Lite                             | <a href="https://arxiv.org/abs/2608.23493">https://arxiv.org/abs/2608.23493</a> · <a href="https://github.com/Galleons2029/SRPO">https://github.com/Galleons2029/SRPO</a>       |
| 2 | ContextPilot: Teaching Agents for Proactive Context Management via Fine-grained RL | 2026-08-28 | Qwen3-8B, Qwen3-14B                                                        | Coarse-grained → action-level credit assignment over branched trajectories during agentic RL                         | InfBench (InfiniteBench), NovelQA, LongMemEval, BrowseComp+            | <a href="https://arxiv.org/abs/2608.28476">https://arxiv.org/abs/2608.28476</a> · <a href="https://github.com/Tencent/ContextPilot">https://github.com/Tencent/ContextPilot</a> |
| 3 | SkillGate: Training In-Policy Skill Selection in Long-Horizon Agents               | 2026-08-21 | Qwen3.5-9B (SFT checkpoint, RL init)                                       | "Selector credit starvation": partitions token support into two credit channels (outcome vs. action-local advantage) | 5 agentic benchmarks incl. SkillsBench, SWE (16-candidate skill slate) | <a href="https://arxiv.org/abs/2608.18852">https://arxiv.org/abs/2608.18852</a>                                                                                                 |
| 4 | CIPO: Contextual Information Policy Optimization for Search Agents                 | 2026-08-06 | Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct                                   | Dense turn-level credit (EALR) to evidence-using reasoning actions, combined with outcome reward                     | HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle + 3 OOD                  | <a href="https://arxiv.org/abs/2608.06128">https://arxiv.org/abs/2608.06128</a>                                                                                                 |
| 5 | MoRSE: Task-Oriented Multi-Agent System with Mixture of Role-Subtask Experts       | 2026-08-10 | Qwen3-4B-Instruct (one of three backbones; also Llama-3.1-8B, Gemma-4-31B) | Hierarchical GRPO with two-layer credit assignment (isolates expert vs. routing quality)                             | Code-generation benchmarks (multi-agent), held-out task domains        | <a href="https://arxiv.org/abs/2608.09251">https://arxiv.org/abs/2608.09251</a>                                                                                                 |

Figure 9: Summary table of recent work on credit assignment in agentic training, assembled during the session.

Figure 10 shows a comparison table from an earlier benchmark-design effort by one contributor. It covers ten related benchmarks across six design dimensions, with each cell read from a source paper. The contributor reported that assembling it took more effort than any part of that project except producing the benchmark data. Benchmark Radar retrieves candidates for this kind of comparison and exposes the matching words and fields for inspection; the comparison itself requires reading the source evidence.



| Bench Name                          | End-to-End Tasks | Fine-Grained Eval | Researcher-Quality Eval | Data Generation  | Multi-Harness Eval | #Tasks |
|-------------------------------------|------------------|-------------------|-------------------------|------------------|--------------------|--------|
| MLE-Bench (Chan et al. 2025)        | ✗                | ✓                 | ✗                       | Transfer&Compose | ✓                  | 75     |
| MLGym-Bench (Nathani et al. 2025)   | ✗                | ✓                 | ✗                       | Automatic        | ✗                  | 13     |
| EXP-Bench (Kon et al. 2025)         | ✓                | ✗                 | ✗                       | Automatic        | ✓                  | 461    |
| ResearchCodeBench (Hua et al. 2026) | ✗                | ✓                 | ✗                       | Transfer&Compose | ✗                  | 212    |
| MLR-Bench (Chen et al. 2026)        | ✓                | ✗                 | ✗                       | Automatic        | ✓                  | 201    |
| PaperBench (Starace et al. 2025)    | ✗                | ✓                 | ✗                       | Transfer&Compose | ✗                  | 8316   |
| AstaBench (Bragg et al. 2025)       | ✓                | ✓                 | ✗                       | Transfer&Compose | ✓                  | 2400+  |
| InnovatorBench (Wu et al. 2025)     | ✓                | ✗                 | ✗                       | Transfer&Compose | ✗                  | 20     |
| AIRS-Bench (Lupidi et al. 2026)     | ✗                | ✓                 | ✗                       | Automatic        | ✓                  | 20     |
| COMPOSITE-Stem (Waters et al. 2026) | ✓                | ✓                 | ✗                       | Manual           | ✗                  | 70     |
| ScienceBoard (Sun et al. 2025)      | ✗                | ✓                 | ✗                       | Manual           | ✗                  | 169    |

Figure 10: A prior-art comparison table assembled manually for an earlier benchmark-design effort. Each row is one related benchmark and each column one design dimension, all read from the source papers by hand.

This case records one workflow combining Benchmark Radar with web search; it has no measured time or accuracy comparison.<sup>3</sup> The summary table and session evidence are available in the project’s public [issue #492](#).

## C Reproducibility, Access, and Citation

This paper evaluates Benchmark Radar v0.10.0 at Git commit 76a6180 with a data cutoff of 2026-09-06. The data build combines dated collection snapshots, benchmark registry records, and model reports, then classifies tracks and packages the results with checksums. Installed clients validate the checksum and record format before activating an update. The evaluation used a clean data rebuild with linting, formatting checks, catalog normalization, classification, release construction, and the full test suite; all 1,236 tests passed. The manuscript is authored in LaTeX, separately from the data generators. The project provides a DOI and a Citation File Format record for citation [6, 18, 35].

| Artifact                      | Canonical or permanent location                                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Archived paper (v0.9.0)       | <a href="https://doi.org/10.5281/zenodo.22167102">https://doi.org/10.5281/zenodo.22167102</a>                                                   |
| Source code                   | <a href="https://github.com/ktwu01/benchmark-radar">https://github.com/ktwu01/benchmark-radar</a>                                               |
| Dashboard                     | <a href="https://benchmark-radar.org/">https://benchmark-radar.org/</a>                                                                         |
| Discovery observations (JSON) | <a href="https://benchmark-radar.org/data/radar.json">https://benchmark-radar.org/data/radar.json</a>                                           |
| Benchmark catalog             | <a href="https://benchmark-radar.org/data/benchmark-index.json">https://benchmark-radar.org/data/benchmark-index.json</a>                       |
| RSS                           | <a href="https://benchmark-radar.org/feed.xml">https://benchmark-radar.org/feed.xml</a>                                                         |
| Citation metadata             | <a href="https://github.com/ktwu01/benchmark-radar/blob/main/CITATION.cff">https://github.com/ktwu01/benchmark-radar/blob/main/CITATION.cff</a> |

Table 8: Access and citation artifacts for Benchmark Radar.

The reported counts were computed from the cumulative discovery data, benchmark catalog, model registry, model-report and score archives, normalized catalog records, and collection configuration. Their repository locations are `site/data/radar.json`, `site/data/benchmark-index.json`, `site/data/models.json`, `data/model_cards.yml`, `data/benchmark_scores.yml`, `data/catalog/`, and `config.yml`. The dated numerical inputs used by the manuscript and figures are stored in `figure-data.tex` beside the LaTeX source.

<sup>3</sup> Jiayu Wang contributed this worked use case and its supporting evidence.

Tables 4 and 5 use the model-report and score archives at commit 76a6180. For the coverage gaps, take each benchmark’s latest score date and latest model-report mention date, then retain gaps of at least 180 days. For headroom, select the highest numeric score in each model-report track with a percentage scale and retain values at least 95. Each selected observation retains its model, source, instrument, and protocol.

To reproduce the data build from a clean checkout, install the project and run `benchmark-radar normalize-catalog`, `benchmark-radar classify`, and `benchmark-radar build-data-release`, in that order. Run `make` in `docs/technical-report/latex/` to build this manuscript from its dated numerical inputs. Rebuilding the data at a later revision does not update those inputs automatically.

The DOI in Table 8 identifies the archived v0.9.0 paper. This manuscript describes v0.10.0 at the cutoff above. The dashboard continues to change; citations of its live counts need a retrieval date.